



UNIVERSIDAD
POLITÉCNICA
DE MADRID



NR sensing testbed & Semantic Compression

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1 Semantic Encoding

2 NR sensing testbed

3 Semantic Compression

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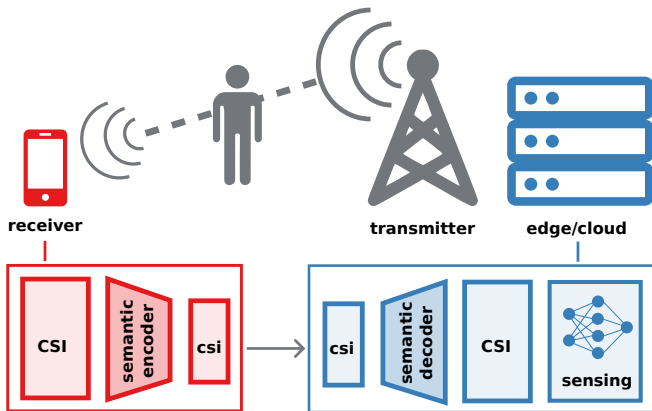


Figura 1: The 5G CSI is compressed through a semantic encoder. The encoded CSI is decoded and processed at the edge/cloud for sensing purposes.

Semantic communications:

transmit necessary information relevant to the specific task

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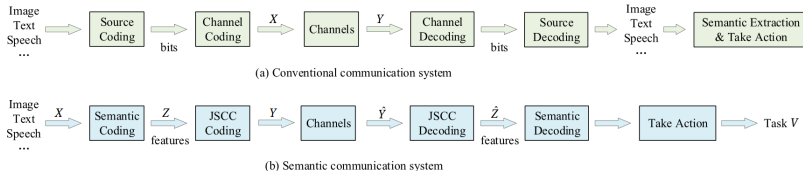


Figura 2: arXiv 2201.01389

What is the SoA doing?

- Semantic Encoders: VAE, codebooks, detection on latent space
- Joint Source Channel Coding as DeepJSCC

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The **pitfall**:

- fixed codebooks/compression rates of size K
- different NN architectures per compression size K
- retrain for each compression size K
- sensing detectors retraining for each compression size K

What do we propose?

- semantic encoder with *on the fly* compression rate K
- segment the grid
- maximum compression outside the SSB
- *reuse* pre-trained sensing detectors

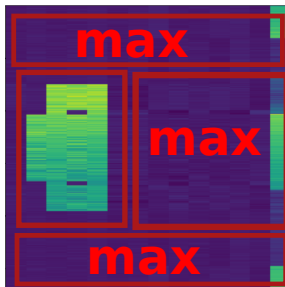


Figura 3

Metric for the semantic similarity: Wasserstein distance.

Experiment: compression impact on sensing accuracy @12k grids

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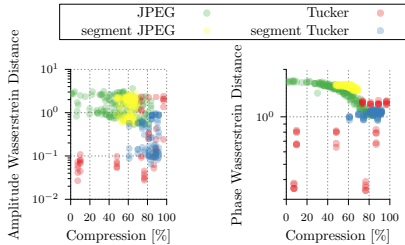


Figura 4

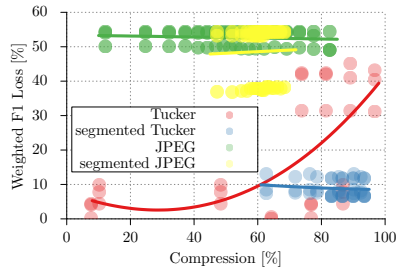


Figura 5

Metric for the semantic similarity: Wasserstein distance.

Experiment: compression impact on sensing accuracy @12k grids

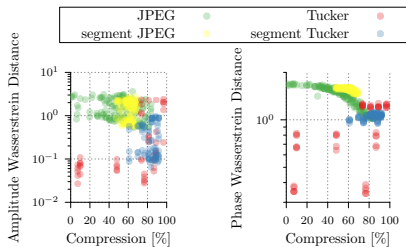


Figura 4

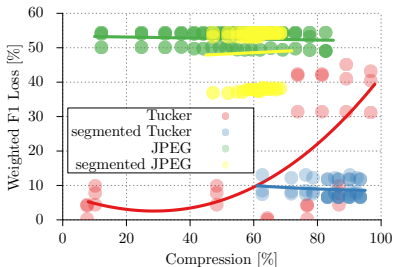


Figura 5

Take-aways:

- more compression doesn't mean more loss
- segmented encoding/compression achieves (i) higher compression; and (ii) better accuracy.

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Tasks done:

- Implemented a person detection script based on *Ultralytics YOLO* to log human presence.
- Implemented relative depth detection with *Depth Anything V2*.
- Conducted preliminary tests in the selected emplacement for the capture campaign.

Person and depth detection for automatic tagging of the data set.

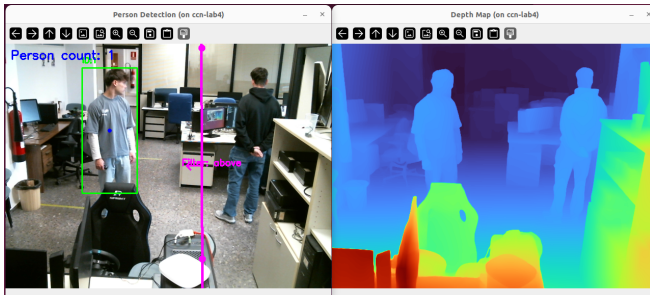


Figura 6: One person detected within virtual boundary

Person and depth detection for automatic tagging of the data set.

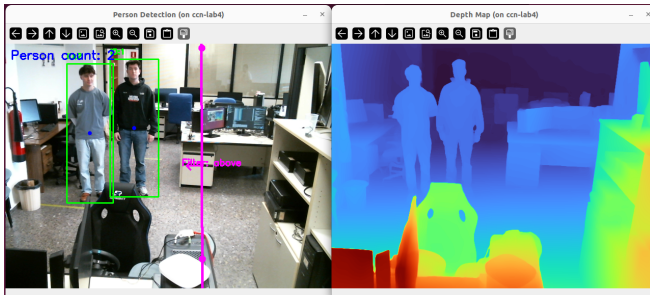


Figura 7: Two people detected within virtual boundary

Pending tasks:

- Implementation, training, and testing of state-of-the-art CNN architectures for presence detection.
- Inference testing with compressed resource grids.
- Capture of a large real scenario dataset.

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After modulation the NR grid is decomposed in amplitude & phase @each RB.

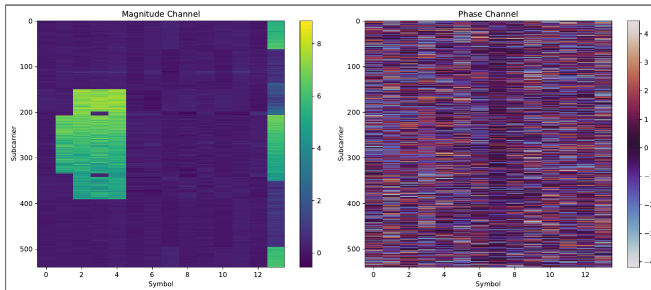


Figura 8

The testbed captures SSBs.

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Problem: each UE exchanges **46 Mbps** using `torch.float64`

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Solutions: to reduce the bandwidth:

- use `torch.float32`, `torch.float16`, etc.; or
- **compress** the NR grid.

We try the following compressions on the NR tensors **grid.pt**:

- Lossless
 - **ZIP** with a 10 % compression
 - **LZ4** with a 6 % compression
- Lossy
 - **JPEG** up to a 98 % compression
 - **JSCC** also ~ 99 % compression
 - **CP** just a ~ 50 % compression
 - **Tucker** up to a 99.44 % compression

We perform compression separately on amplitude/phase channels.

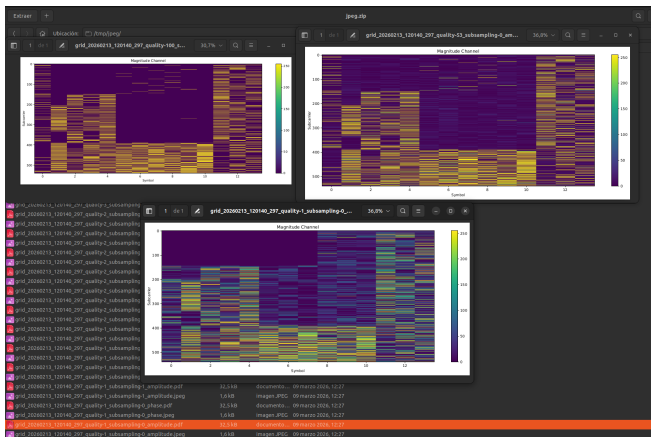


Figura 9: 100, 50, and 1 JPEG quality

Original **NR grid** tensor $T \in F^{n_1 \times n_2 \times n_3}$ decomposed as

$$T = \mathcal{T} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)} \quad (1)$$

with $\mathcal{T} \in F^{d_1 \times d_2 \times d_3}$ the compressed tensor, and the k -mode products:

$$(\mathcal{T} \times_1 U^{(1)})(i_1, j_2, j_3) = \sum_{j_1=1}^{d_1} \mathcal{T}(j_1, j_2, j_3) U^{(1)}(j_1, i_1) \quad (2)$$

$$(\mathcal{T} \times_2 U^{(2)})(j_1, i_2, j_3) = \sum_{j_2=1}^{d_2} \mathcal{T}(j_1, j_2, j_3) U^{(2)}(j_2, i_2) \quad (3)$$

$$(\mathcal{T} \times_3 U^{(3)})(j_1, j_2, i_3) = \sum_{j_3=1}^{d_3} \mathcal{T}(j_1, j_2, j_3) U^{(3)}(j_3, i_3) \quad (4)$$

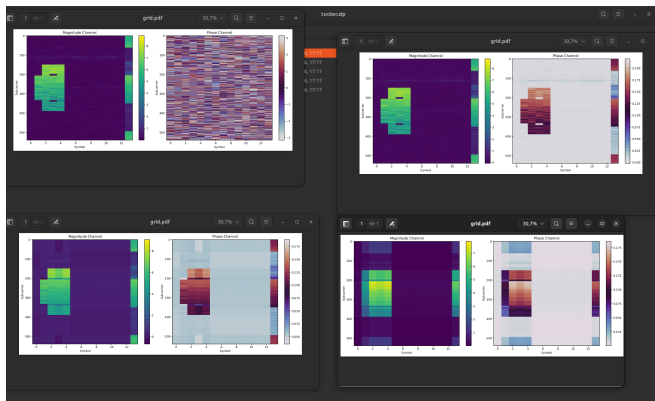


Figure 10: Tucker with $d_1 = 2, 0.50, 0.75, 1.00$ % compression.

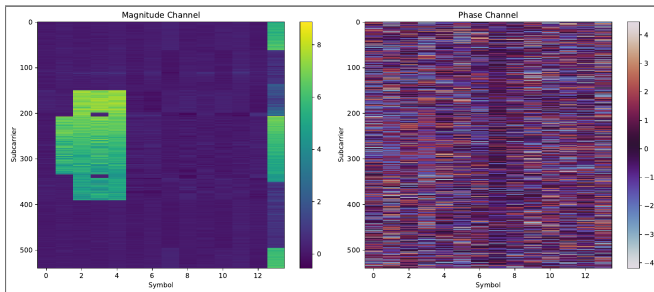


Figura 11: Tucker with $d_1 = 1$ and 50% compression.

- Apply semantic segmentation and compression:
 - Compress SSB
 - Compress x4 regions around SSB
- Assess AI/ML sensing detectors performance with compressed NR grids